

TASK 2: Stock Portfolio Tracker (with Execution Output)

CodeAlpha Python Programming Internship • Project Submission

1. Source Code Implementation

```
def stock_portfolio_tracker():
    stock_prices = {
        "AAPL": 180, "TSLA": 250, "GOOG": 150, "AMZN": 175, "MSFT": 400
    }
    portfolio = {}
    print("--- Stock Portfolio Tracker ---")
    print(f"Available stocks and prices: {stock_prices}\n")

    while True:
        stock_name = input("Enter stock symbol (or type 'done' to finish): ").upper().strip()
        if stock_name == 'DONE':
            break
        if stock_name not in stock_prices:
            print("Stock not found in our database. Please try again.")
            continue
        try:
            quantity = int(input(f"Enter quantity for {stock_name}: "))
            if quantity <= 0:
                print("Quantity must be greater than 0.")
                continue
            portfolio[stock_name] = portfolio.get(stock_name, 0) + quantity
        except ValueError:
            print("Invalid input. Please enter a valid number for quantity.")

    print("\n--- Portfolio Summary ---")
    total_investment = 0
    report_lines = ["--- Stock Portfolio Report ---\n"]

    for stock, qty in portfolio.items():
        price = stock_prices[stock]
        cost = qty * price
        total_investment += cost
        line = f"{stock}: {qty} shares @ ${price} each = ${cost}"
        print(line)
        report_lines.append(line + "\n")

    summary_text = f"\nTotal Portfolio Value: ${total_investment}"
    print(summary_text)
    report_lines.append(summary_text)

    try:
        with open("portfolio_report.txt", "w") as file:
            file.writelines(report_lines)
        print("\nReport successfully saved to 'portfolio_report.txt'")
    except Exception as e:
        print(f"Error saving file: {e}")
```

```
if __name__ == "__main__":  
    stock_portfolio_tracker()
```

2. Expected Terminal Output

```
--- Stock Portfolio Tracker ---  
Available stocks and prices: {'AAPL': 180, 'TSLA': 250, 'GOOG': 150, 'AMZN': 175, 'MSFT': 400}  
  
Enter stock symbol (or type 'done' to finish): aapl  
Enter quantity for AAPL: 5  
Enter stock symbol (or type 'done' to finish): tsla  
Enter quantity for TSLA: 2  
Enter stock symbol (or type 'done' to finish): done  
  
--- Portfolio Summary ---  
AAPL: 5 shares @ $180 each = $900  
TSLA: 2 shares @ $250 each = $500  
  
Total Portfolio Value: $1400  
  
Report successfully saved to 'portfolio_report.txt'
```